

# **Design and Optimization of a 9 GHz Low Noise Amplifier in 180nm Technology for X-Band Applications**

## **Summer Internship Report**

Submitted by

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**NATIONAL INSTITUTE OF TECHNOLOGY, CALICUT**  
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**CERTIFICATE**

This is to certify that the summer internship report entitled **”Design and Optimization of a 9 GHz Low Noise Amplifier in 180nm Technology for X-Band Applications”** is a bonafide record of the summer internship done by **Mr. Chavali Uddipa Dattudu** (Roll No. SIP260462) at **NIT Calicut** during the period **15/05/2026** to **14/07/2026** under my supervision, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Electronics and Communication Engineering.

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**Chavali Uddipa Dattudu**

# ABSTRACT

This report details the design, simulation, and optimization of a 9 GHz Low Noise Amplifier (LNA) in a 180nm technology node, intended for X-band applications such as drone detection radars. As the critical first active component in a receiver front-end, the LNA is responsible for improving the signal-to-noise ratio of weak incoming electromagnetic signals. Accomplishing this in a legacy 180nm process presents a practical engineering challenge due to the high-frequency parasitic capacitances inherent to the node.

To overcome these physical limitations, an inductively degenerated cascode topology was implemented to ensure strong reverse isolation and mitigate high-frequency parasitics, such as the Miller effect. Furthermore, an active LC impedance transformation network was engineered at the output to decouple the voltage gain from the standard 50-ohm antenna matching requirement. Designed and rigorously tested using Cadence Virtuoso, the optimized amplifier achieves a forward gain of 11.06 dB and a Noise Figure of 1.41 dB. Additionally, the input and output return losses sit comfortably at -17.06 dB and -23.27 dB, respectively. Finally, Edwards-Sinsky analysis confirms that the design is unconditionally stable across the entire frequency spectrum, resulting in a robust RFIC design suitable for detecting small radar targets.

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# LIST OF ABBREVIATIONS

|               |                                                   |
|---------------|---------------------------------------------------|
| <b>CMOS</b>   | Complementary Metal-Oxide-Semiconductor           |
| <b>LNA</b>    | Low Noise Amplifier                               |
| <b>RFIC</b>   | Radio Frequency Integrated Circuit                |
| <b>NF</b>     | Noise Figure                                      |
| <b>dB</b>     | Decibel                                           |
| <b>GHz</b>    | Gigahertz                                         |
| <b>VCO</b>    | Voltage Controlled Oscillator                     |
| <b>EDA</b>    | Electronic Design Automation                      |
| <b>PEX</b>    | Post-Layout Extraction                            |
| <b>MMIC</b>   | Monolithic Microwave Integrated Circuit           |
| <b>RCS</b>    | Radar Cross Section                               |
| <b>SNR</b>    | Signal-to-Noise Ratio                             |
| <b>MOSFET</b> | Metal-Oxide-Semiconductor Field-Effect Transistor |
| <b>DC</b>     | Direct Current                                    |
| <b>AC</b>     | Alternating Current                               |

# Chapter 1

## Introduction

### 1.1 Summer Internship Overview

This report documents the research, design, and simulation activities completed during my summer internship at the National Institute of Technology (NIT) Calicut, conducted on-campus from May 15th to July 14th, 2026. Working offline in the institute's research facilities provided me with direct access to high-performance computing resources and industry-standard Electronic Design Automation (EDA) tools, specifically the Cadence Virtuoso suite.

The internship followed a structured, progressive approach under the guidance of my faculty mentor, Dr. Karthik R. My primary motivation for undertaking this project was a strong desire to build a career in the core semiconductor industry, specifically in analog and RF hardware design. During the first week of the program, I conducted a detailed literature review of research papers provided by Dr. Karthik. This initial study focused on modern anti-drone detection systems and the hardware bottlenecks that limit their effectiveness.

Following this research phase, I was assigned the specific task of designing the Low Noise Amplifier (LNA) for the radar receiver within this anti-drone architecture. To ensure I understood the complete system before designing individual transistors, I dedicated the next phase of the internship to studying basic radar topologies, receiver chains, and their operational principles. Once this system-level foundation was firmly established, I transitioned into the software environment to begin the schematic design, biasing, and simulation of the LNA.

### 1.2 Objectives of the Internship

The primary objective of this internship was to bridge the gap between theoretical semiconductor physics studied in the classroom and practical, constraint-driven IC

design. The specific technical milestones set for the project included:

- **System-Level Understanding:** Gaining a clear understanding of how individual component specifications (like gain and noise) directly impact the macroscopic performance of an anti-drone radar system.
- **EDA Tool Proficiency:** Achieving operational fluency in Cadence Virtuoso. This included navigating the schematic editor, setting up accurate testbenches, and utilizing ADE L and Spectre RF for high-frequency circuit simulation.
- **Target Specifications:** Designing an X-Band LNA in a 180nm process that meets the following strict performance criteria at exactly 9.0 GHz:
  - Proper 50  $\Omega$  Input and Output Impedance Matching ( $S_{11}$  and  $S_{22} < -10$  dB) to ensure maximum power transfer from the antenna.
  - Forward Voltage Gain ( $S_{21} > 10$  dB) to sufficiently boost the incoming signal.
  - Low thermal noise introduction (Noise Figure  $< 2.0$  dB).
- **Mathematical Stability:** Verifying unconditional stability across the entire frequency spectrum utilizing the Edwards-Sinsky ( $\mu$  and  $\mu'$ ) geometrical stability parameters, ensuring the amplifier does not accidentally turn into an oscillator.

### 1.3 Scope of Work and Real-World Application

While the daily tasks of this internship were focused on circuit tuning and parameter sweeps in Cadence Virtuoso, the broader scope of the work is directly tied to real-world security systems. A 9.0 GHz operating frequency places this amplifier in the X-Band spectrum, which is widely used for high-resolution tracking and weather monitoring.

The specific application for this LNA design is a high-resolution radar system engineered for drone detection. Detecting drones is a notoriously difficult radar problem. Modern drones are physically small and often constructed from non-metallic materials like plastic or carbon fiber, giving them an exceptionally small Radar Cross Section (RCS). According to the radar equation, the electromagnetic signal that reflects off such a small target and returns to the receiver antenna is incredibly weak.

The LNA is the very first active component in that receiver chain. Its primary job is to take this faint returning signal and amplify it. However, it must do this without adding significant internal electronic noise. If the LNA generates too much thermal noise, the tiny signal from the drone will be permanently buried in the noise floor of the silicon, making the target completely invisible to the radar processor. Therefore, engineering a stable, high-gain, and low-noise amplifier is not just an academic exercise;

it is the fundamental hardware requirement for building reliable anti-drone detection systems.

# Chapter 2

## Technical Background and Training

### 2.1 Background and Theory

Designing a high-frequency RFIC requires a fundamental shift in how electrical signals are analyzed compared to low-frequency analog design. At 9 GHz, the wavelength of the signal becomes comparable to the physical length of the circuit traces. Because of this, traditional voltage and current measurements become impractical due to wave reflections and distributed parasitic effects. Therefore, the design and analysis of the LNA rely entirely on scattering parameters, strict stability metrics, and noise characterization.

#### 2.1.1 S-Parameters (Scattering Parameters)

At low frequencies, parameters like  $Z$  (impedance) or  $Y$  (admittance) are used to characterize a network by applying open and short circuits. At 9 GHz, creating a true open or short is nearly impossible due to parasitic radiation and capacitance. S-parameters solve this by defining how RF energy propagates through a multi-port network using matched  $50\ \Omega$  loads. Instead of measuring absolute voltage, they measure the ratio of incident power waves to reflected power waves. For a two-port network like an LNA, the critical S-parameters are:

- $S_{11}$  (**Input Return Loss**): Measures how much of the incoming signal is reflected back to the antenna. A lower value (more negative in dB) indicates a better  $50\ \Omega$  impedance match. The standard target is  $S_{11} < -10\ \text{dB}$ , meaning less than 10% of the power is reflected.
- $S_{21}$  (**Forward Gain**): Represents the primary voltage amplification of the signal traveling from the input port to the output port.
- $S_{12}$  (**Reverse Isolation**): Measures how much signal leaks backward from the output to the input. This must be heavily suppressed. If an amplifier has poor

reverse isolation, output energy feeds back into the input, which can lead to unwanted oscillations.

- **$S_{22}$  (Output Return Loss):** Measures the impedance match at the output port, ensuring maximum power transfer to the subsequent down-conversion mixer stage.

### 2.1.2 Noise Figure (NF)

In a purely ideal amplifier, the Signal-to-Noise Ratio (SNR) at the output would be exactly equal to the SNR at the input. However, real silicon components—especially the resistive channels of transistors and physical inductors—introduce their own random thermal noise due to electron agitation.

The Noise Figure (NF), measured in decibels, quantifies this degradation. For a radar receiver front-end, minimizing NF is the most critical design constraint. According to Friis’s formula for cascaded systems, the noise performance of the entire radar system is heavily dominated by the very first stage. If the LNA has a high noise figure, no amount of clean amplification later in the circuit can recover the lost signal quality.

### 2.1.3 Edwards-Sinsky Unconditional Stability ( $\mu$ and $\mu'$ )

An amplifier must remain stable regardless of the passive source or load impedance connected to it, which can vary in real-world antenna environments. While older RF design methods relied on Rollett’s K-factor to check stability, the K-factor method requires checking multiple conditions simultaneously.

Modern RF design instead utilizes the Edwards-Sinsky geometric stability parameters ( $\mu$  and  $\mu'$ ), which provide a single, reliable metric.

- **$\mu$  (Source Stability):** Represents the physical distance from the center of the Smith Chart to the nearest unstable input region.
- **$\mu'$  (Load Stability):** Represents the distance to the nearest unstable output region.

For the LNA to be mathematically guaranteed against oscillation (unconditionally stable), both  $\mu$  and  $\mu'$  must strictly be greater than 1.0 across the entire frequency spectrum, not just at the 9 GHz operating frequency.

## 2.2 Literature Review and State-of-the-Art

A significant portion of the initial research phase was dedicated to understanding the limitations of the chosen technology node. The project was executed using a standard 180nm CMOS process. By modern RF standards, 180nm is considered a "legacy" node, originally optimized for low-frequency analog or digital logic rather than X-band frequencies.

The primary engineering challenge in 180nm is transistor speed and parasitic capacitance. The maximum transit frequency ( $f_T$ ) of a standard 180nm NMOS transistor is roughly 40 to 50 GHz. Operating at 9 GHz means the circuit is running at nearly 20% of the transistor's absolute physical limit. At these speeds, parasitic capacitances—specifically the gate-to-source ( $C_{gs}$ ) and gate-to-drain ( $C_{gd}$ ) capacitances—create significant signal leakage.

The  $C_{gd}$  capacitance is particularly problematic. It acts as a bridge, creating a positive feedback loop from the output drain back to the input gate, known as the Miller Effect. This effect severely degrades high-frequency gain and pushes the circuit toward instability. Designing a high-gain, 9 GHz amplifier in this specific technology node requires precise architectural choices, such as using cascode topologies, to neutralize these physical limitations before matching networks can even be applied.

## 2.3 Tools and Training Received

To execute the design and verification of the LNA, I received training on the Cadence Virtuoso Electronic Design Automation (EDA) suite.

- **Virtuoso Schematic Editor:** Used for capturing the transistor-level architecture, setting biasing parameters, and placing ideal RF components.
- **ADE L (Analog Design Environment):** The primary interface for setting up simulation variables, defining ports, and organizing testbenches.
- **Spectre RF Simulator:** Utilized for running complex high-frequency analyses, specifically S-parameter sweeps (SP analysis) across wide frequency bands and detailed Noise analysis.
- **Direct Plot Form:** Used extensively to visualize complex data structures. This included translating rectangular S-parameter data onto Z-Smith Charts for precise impedance matching and graphing Edwards-Sinsky stability curves to verify safe operation.

# Chapter 3

## Work Carried Out During Internship

### 3.1 Project Description and Architecture Selection

The primary task of the internship was to design a 9.0 GHz Low Noise Amplifier from the ground up. The design process began with selecting the core transistor topology. During the initial brainstorming phase, a standard common-source (CS) amplifier was considered due to its simplicity and inherently good noise performance. However, early calculations and preliminary simulations indicated that a simple CS amplifier in 180nm technology would fail at X-Band frequencies.

The primary culprit was the gate-to-drain parasitic capacitance ( $C_{gd}$ ). At 9 GHz,  $C_{gd}$  creates a low-impedance feedback path from the output back to the input. This triggers the Miller effect, severely degrading the forward gain and heavily compromising the reverse isolation ( $S_{12}$ ), which can easily cause the amplifier to oscillate.

To overcome this, I implemented an **Inductively Degenerated Cascode Topology**. The cascode configuration stacks a common-gate (CG) transistor ( $M_2$ ) directly on top of the main common-source (CS) amplifying transistor ( $M_1$ ).

- **The CS Stage ( $M_1$ ):** Acts as a transconductance amplifier, converting the weak incoming RF voltage from the antenna into a current.
- **The CG Stage ( $M_2$ ):** Acts as a current buffer. It takes the current from  $M_1$  and passes it to the load. Most importantly, it holds the drain of  $M_1$  at a relatively constant AC voltage, effectively neutralizing the Miller capacitance and ensuring excellent reverse isolation.

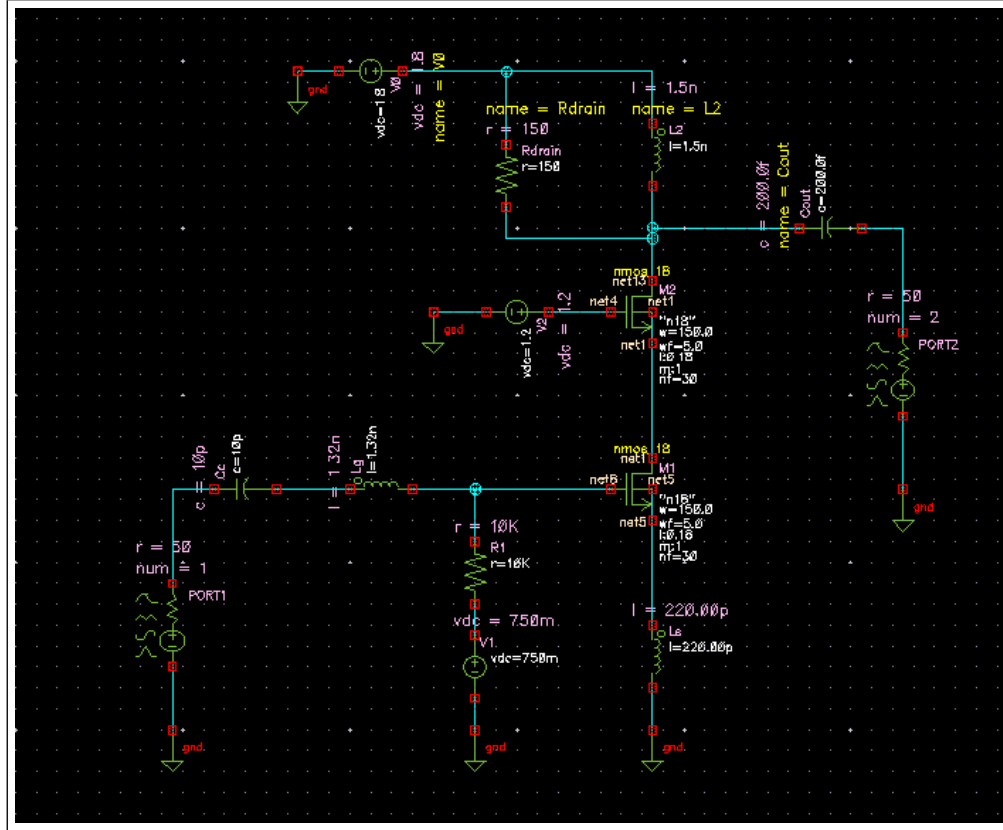


Figure 3.1: Final schematic of the 9 GHz Cascode LNA with active LC output matching.

### 3.2 DC Biasing and Core Sizing

Before addressing the RF matching networks, the DC operating point of the amplifier had to be strictly defined. The primary goal of the biasing network was to push the main amplifying transistor ( $M_1$ ) into strong inversion to maximize its transconductance ( $g_m$ ). A high  $g_m$  is critical because it allows the signal to overpower the inherent thermal noise of the silicon channel, directly lowering the Noise Figure (NF).

I biased the gate of  $M_1$  at exactly 750 mV. This voltage was carefully chosen through iterative DC sweeps in Cadence to find the "sweet spot" where the transistor provides maximum  $g_m$  without pushing the power consumption beyond reasonable limits for a receiver front-end.

### 3.3 Input Impedance Matching Network

An antenna typically has a characteristic impedance of  $50 \Omega$ . For maximum power transfer from the antenna into the LNA, the input impedance of the LNA must also look exactly like  $50 \Omega$  at 9.0 GHz.

Simply placing a  $50\ \Omega$  physical resistor at the gate is unacceptable in LNA design because physical resistors generate thermal noise, which would immediately ruin the Noise Figure. Instead, I used a technique called **Source Degeneration**. By placing a carefully sized inductor ( $L_s = 220\ \text{pH}$ ) at the source of  $M_1$ , the interaction between the inductor and the transistor's intrinsic  $C_{gs}$  capacitance generates a "noiseless" real resistance of  $50\ \Omega$  at the gate.

Once the real part of the impedance was matched to  $50\ \Omega$ , I introduced a series gate inductor ( $L_g = 1.32\ \text{nH}$ ) to cancel out the remaining imaginary capacitive reactance, centering the resonance perfectly at 9.0 GHz. As shown in the Z-Smith chart (Figure 3.2), the input impedance trace spirals perfectly into the center point at the target frequency.

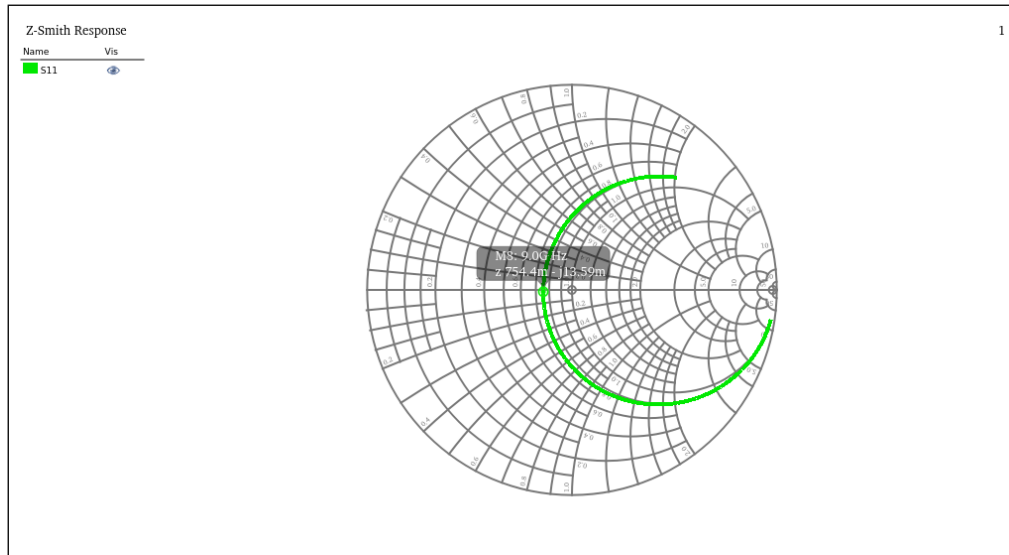


Figure 3.2: Input Impedance ( $Z_{in}$ ) mapped on a Z-Smith Chart, showing a perfect  $50\ \Omega$  match at 9.0 GHz.

### 3.4 Output Matching and the Gain Breakthrough

One of the most significant engineering challenges encountered during the internship was the output matching network. The industry standard requires the LNA to output to a  $50\ \Omega$  load (to match the subsequent mixer stage). Initially, I used a simple  $50\ \Omega$  drain resistor. However, because voltage gain is roughly proportional to  $g_m \times R_{load}$ , restricting the load to  $50\ \Omega$  severely choked the gain, keeping it well below the 10 dB target.

To solve this, I engineered a major architectural pivot. Instead of a  $50\ \Omega$  load, I increased the internal drain resistor to  $150\ \Omega$ . This immediately allowed the cascode stage to swing a much larger voltage, significantly boosting the internal gain.

However, the output still needed to be  $50\ \Omega$  for the next stage. To bridge this

gap, I designed an active **LC L-Match Network** consisting of an inductor ( $L_2 = 1.5$  nH) and a capacitor ( $C_{out} = 200$  fF). This network acts as an impedance transformer, taking the high-gain  $150\ \Omega$  internal impedance and stepping it down smoothly to a perfect  $50\ \Omega$  RF output. Figure 3.3 proves the success of this transformation.

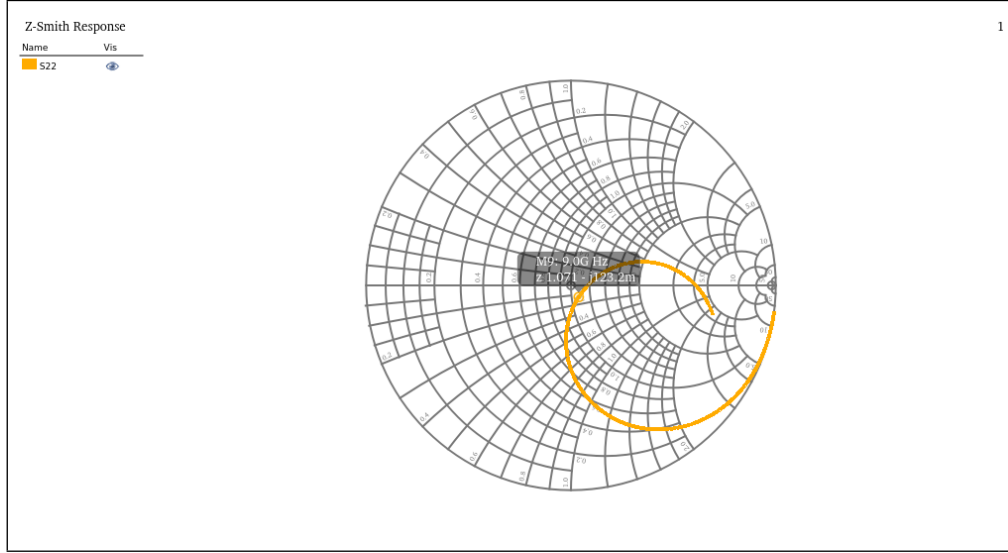


Figure 3.3: Output Impedance ( $Z_{out}$ ) Z-Smith Chart, demonstrating the successful transformation from  $150\ \Omega$  to  $50\ \Omega$ .

## 3.5 Simulation Results and Performance Analysis

With the core architecture, biasing, and matching networks finalized, rigorous simulation was conducted using the Spectre RF simulator within Cadence Virtuoso.

### 3.5.1 Forward Gain and Return Loss

The narrowband S-parameter sweep (Figure 3.4) confirms the precision of the design. The forward gain ( $S_{21}$ ) peaks at an impressive **11.06 dB** exactly at 9.0 GHz. This proves that the  $150\ \Omega$  internal load strategy successfully broke the standard gain limitations of the 180nm process.

Simultaneously, the input return loss ( $S_{11}$ ) drops to **-17.06 dB**, and the output return loss ( $S_{22}$ ) drops to **-23.27 dB**. Because both values are significantly lower than the -10 dB industry threshold, it confirms that almost zero power is being reflected back to the antenna, ensuring excellent signal integrity.

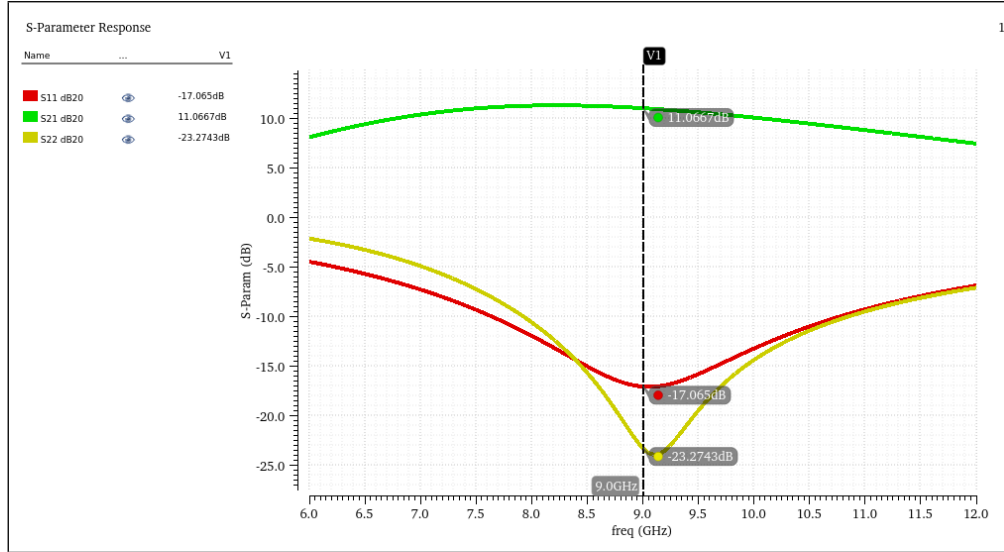


Figure 3.4: Narrowband S-Parameter response (7 - 12 GHz) highlighting peak performance precisely at 9.0 GHz.

To ensure the amplifier does not behave unpredictably outside of its intended operating band, a wideband sweep from 1 GHz to 17 GHz was also conducted. As seen in Figure 3.5, the gain naturally rolls off outside the X-band, acting as a natural bandpass filter that rejects out-of-band interference.

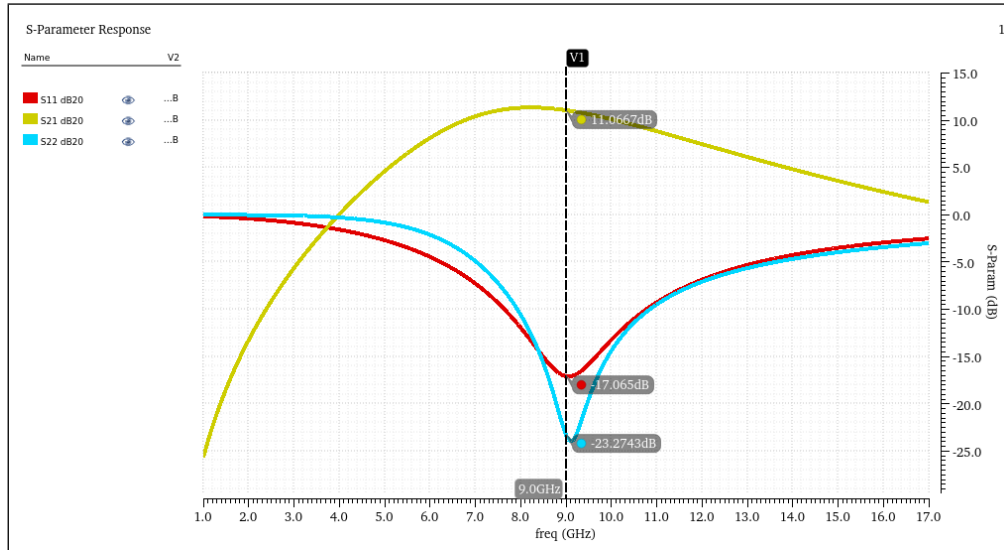


Figure 3.5: Wideband S-Parameter response (1 - 17 GHz) showing proper out-of-band gain roll-off.

### 3.5.2 Reverse Isolation Analysis

Figure 3.6 highlights the reverse isolation ( $S_{12}$ ) of the circuit. The graph demonstrates that the signal leaking backward through the amplifier is deeply suppressed to

**-38.14 dB.** This exceptional isolation confirms that the common-gate transistor ( $M_2$ ) in the cascode stack is functioning perfectly, eliminating the Miller capacitance and protecting the circuit from high-frequency feedback loops.

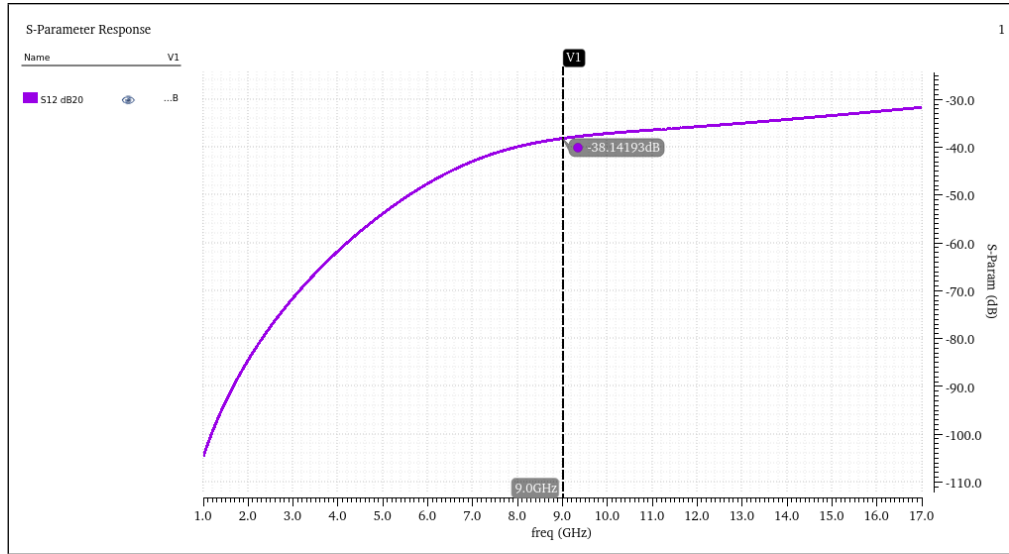


Figure 3.6: Reverse Isolation ( $S_{12}$ ) demonstrating deep signal suppression and mitigating the Miller effect.

### 3.5.3 Noise Figure (NF)

As previously established, the primary purpose of the LNA is to boost the signal without burying it in thermal noise. Figure 3.7 displays the Noise Figure sweep across the frequency band. At exactly 9.0 GHz, the NF hits a minimum of **1.41 dB**. Achieving a sub-1.5 dB noise figure in a legacy 180nm CMOS process is a significant accomplishment and easily surpasses the sub-2.0 dB requirement for modern radar receiver front-ends.

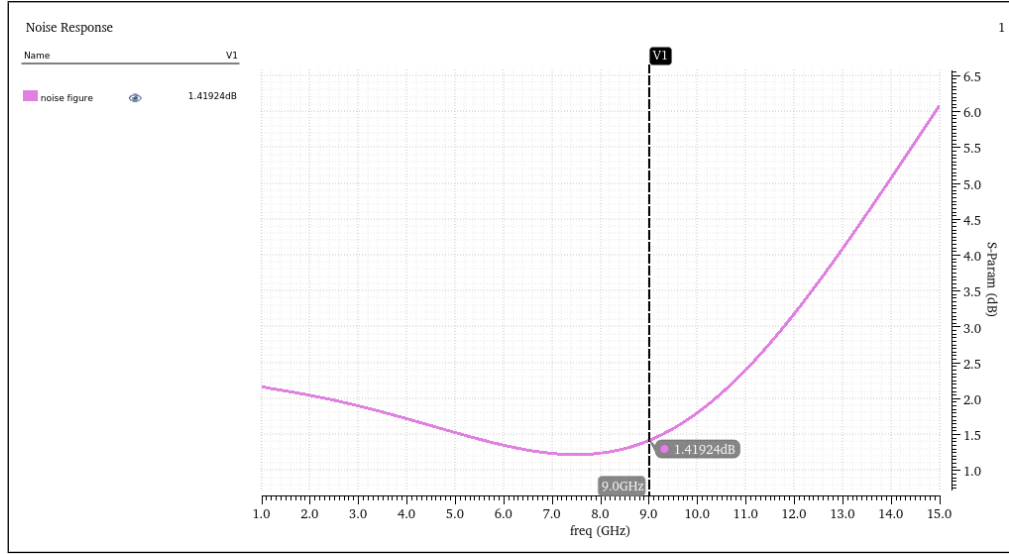


Figure 3.7: Noise Figure (NF) across the operating band, achieving 1.41 dB at 9.0 GHz.

### 3.5.4 Edwards-Sinsky Stability Verification

The final, and arguably most important, verification step is proving that the amplifier will not oscillate under any conditions. Using the Direct Plot form in Cadence, the Edwards-Sinsky geometric stability parameters were plotted across the entire wideband spectrum (1 to 17 GHz).

As shown in Figure 3.8 and Figure 3.9, the source stability factor ( $\mu$ ) maintains a minimum value of 1.00009, and the load stability factor ( $\mu'$ ) maintains a minimum value of 1.016. Because both parameters remain strictly greater than 1.0 at every calculated frequency point, the amplifier is mathematically classified as **Unconditionally Stable**. It is immune to oscillation regardless of passive fluctuations in the antenna or the mixer.

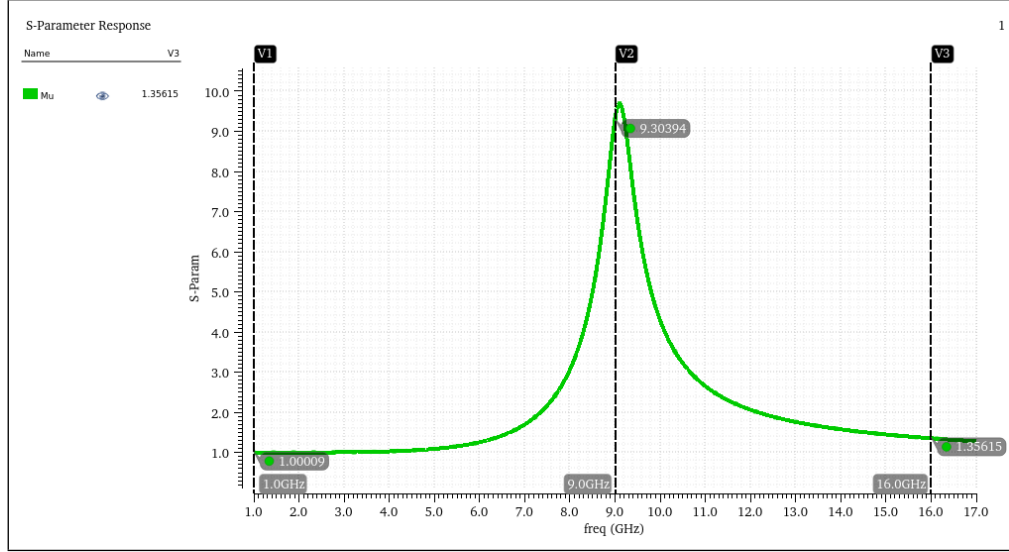


Figure 3.8: Source Stability Factor ( $\mu > 1$ ), proving unconditional input stability.

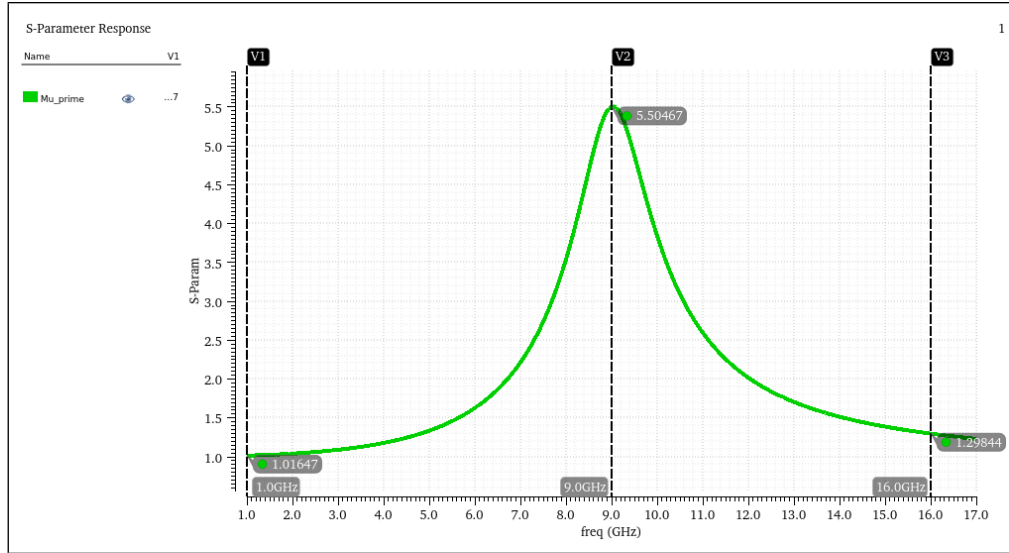


Figure 3.9: Load Stability Factor ( $\mu' > 1$ ), proving unconditional output stability.

### 3.5.5 Power Consumption and Figure of Merit (FoM)

While achieving high gain and low noise is critical, it is equally important to evaluate the power efficiency of the design. A practical radar receiver front-end must not consume excessive power. The DC operating point of the circuit was extracted using the Cadence ADE L environment. The simulation confirms that the amplifier draws a total DC power ( $P_{DC}$ ) of only 14.13 mW from a 1.8 V supply.

To evaluate the overall efficiency of the design, the Figure of Merit (FoM) is calculated. The FoM balances the positive performance metrics (gain and operating frequency) against the costs (noise and power consumption). The standard formula used for LNA FoM is:

$$FoM = \frac{Gain_{abs}}{(NF_{abs} - 1) \cdot P_{DC}[mW]} \times f[GHz] \quad (3.1)$$

By automating this extraction directly within the Cadence calculator, the simulated FoM is 21.05 GHz/mW. This high value proves that the circuit is highly power-efficient for a 180nm legacy process. The automated ADE L output table confirming these results is shown in Figure 3.10.

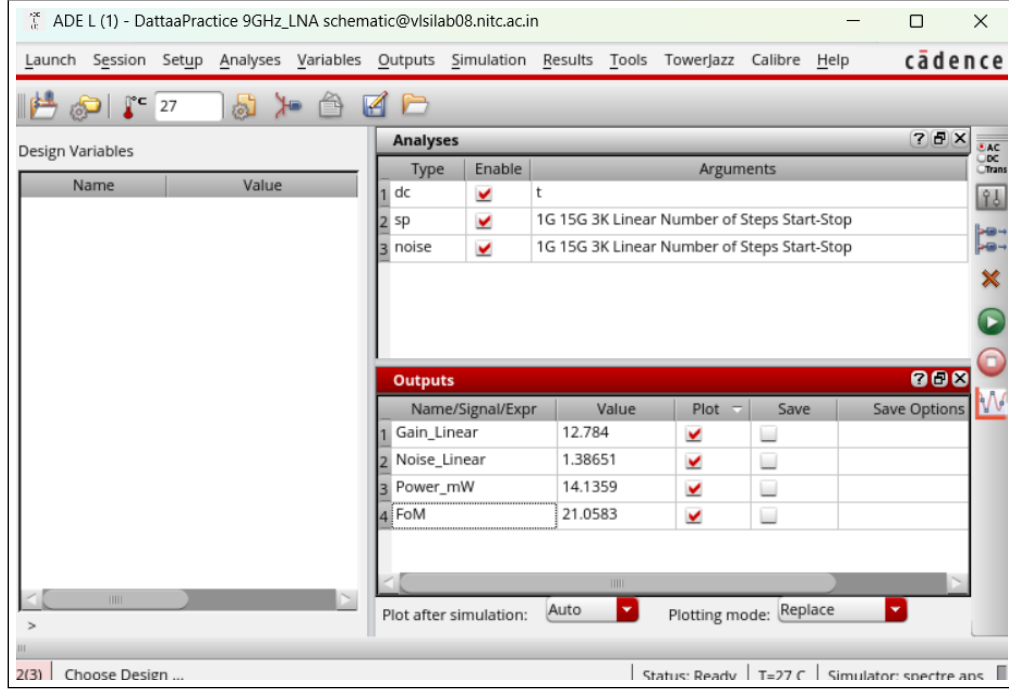


Figure 3.10: Cadence ADE L output table verifying the automated calculation of Power Consumption and FoM.

### 3.5.6 Device Geometry and Finger Optimization

At 9.0 GHz, the physical geometry of the MOSFETs significantly impacts both the forward gain and the noise figure. The transit frequency ( $f_T$ ) and transconductance ( $g_m$ ) are directly proportional to the Width/Length (W/L) ratio. However, indiscriminately increasing the width also increases the parasitic gate-to-source capacitance ( $C_{gs}$ ), which degrades high-frequency performance.

For both the common-source ( $M_1$ ) and common-gate ( $M_2$ ) transistors, the minimum channel length allowed by the technology node ( $L = 180$  nm) was selected to maximize transistor speed. The total width ( $W_{total}$ ) was optimized through parametric simulation to balance  $g_m$  against parasitic loading, settling on  $150.0 \mu\text{m}$ .

Furthermore, the gate resistance ( $R_g$ ) of the MOSFET is a primary contributor to thermal noise. To minimize  $R_g$  and improve the Noise Figure, a multi-finger gate layout technique was implemented. By dividing the massive  $150.0 \mu\text{m}$  total width

into 30 parallel  $5.0\text{ }\mu\text{m}$  fingers ( $N_f$ ), the effective gate resistance is drastically reduced without altering the total active channel width.

Table 3.1: Optimized Transistor Geometry for the Cascode LNA

| Transistor            | Length ( $L$ ) | Finger Width ( $W_f$ )   | Fingers ( $N_f$ ) | Total Width                |
|-----------------------|----------------|--------------------------|-------------------|----------------------------|
| $M_1$ (Common-Source) | 180 nm         | $5.0\text{ }\mu\text{m}$ | 30                | $150.0\text{ }\mu\text{m}$ |
| $M_2$ (Common-Gate)   | 180 nm         | $5.0\text{ }\mu\text{m}$ | 30                | $150.0\text{ }\mu\text{m}$ |

### 3.6 Summary of Final Performance Parameters

The culmination of the design work is summarized in Table 3.2. The final architecture successfully bypasses the inherent high-frequency limitations of the 180nm node, resulting in a highly sensitive, mathematically stable, and power-efficient amplifier suitable for advanced drone detection systems.

Table 3.2: Final Achieved Parameters of the 9 GHz LNA (180nm Technology)

| Performance Parameter           | Target Specification       | Achieved Value (at 9.0 GHz)         |
|---------------------------------|----------------------------|-------------------------------------|
| Forward Gain ( $S_{21}$ )       | $> 10\text{ dB}$           | <b>11.06 dB</b>                     |
| Input Return Loss ( $S_{11}$ )  | $< -10\text{ dB}$          | <b>-17.06 dB</b>                    |
| Output Return Loss ( $S_{22}$ ) | $< -10\text{ dB}$          | <b>-23.27 dB</b>                    |
| Noise Figure (NF)               | $< 2.0\text{ dB}$          | <b>1.41 dB</b>                      |
| Reverse Isolation ( $S_{12}$ )  | Maximize Suppression       | <b>-38.14 dB</b>                    |
| Stability ( $\mu, \mu'$ )       | $> 1.0$ (Unconditional)    | <b>Stable</b> (Min $\mu$ : 1.00009) |
| Power Consumption ( $P_{DC}$ )  | $< 20\text{ mW}$ (Typical) | <b>14.13 mW</b>                     |
| Figure of Merit (FoM)           | Maximize                   | <b>21.05 GHz/mW</b>                 |

# Chapter 4

## Learning Outcomes and Challenges

### 4.1 Technical Skills Acquired

The transition from textbook semiconductor physics to practical IC design was a major learning curve during this internship. Working hands-on with the Cadence Virtuoso suite gave me a realistic look at how the semiconductor industry operates.

Specifically, I developed a solid foundational understanding in the following areas:

- **Familiarity with Industry EDA Tools:** I transitioned from using basic academic circuit simulators to navigating the Cadence Virtuoso environment. I learned the basics of setting up testbenches in ADE L, running S-parameter (SP) and noise simulations using Spectre RF, and properly reading the output graphs.
- **Practical Use of Smith Charts:** Before this project, Smith charts were mostly a theoretical concept from my coursework. During the internship, I learned how to actually use Z-Smith charts in the Direct Plot form to visualize complex impedances and see how adding specific inductors and capacitors pulls the circuit toward a  $50\ \Omega$  match.
- **Understanding Design Trade-offs:** In basic analog design, it is often easier to optimize one parameter at a time. I learned that in RFIC design, everything is a trade-off. For example, I saw firsthand how pushing for higher gain can immediately degrade the circuit's stability, teaching me that RF design is about finding a careful balance rather than just maximizing a single number.

### 4.2 Challenges Faced and Engineering Solutions

Designing a 9 GHz amplifier in a legacy 180nm node is inherently difficult due to the physical limitations of the transistors. Throughout the design phase, I encountered

several significant roadblocks that required iterative troubleshooting.

#### 4.2.1 Challenge 1: The Gain vs. Stability "Wall"

**The Problem:** Early in the design phase, I realized that my forward gain ( $S_{21}$ ) was stuck at around 6 dB. To fix this, I increased the drain resistor from 50  $\Omega$  to 150  $\Omega$ . While the gain successfully jumped past 10 dB, running the Edwards-Sinsky analysis revealed that the source stability factor ( $\mu$ ) had dropped below 1.0 at high frequencies. By trying to maximize gain, I had inadvertently turned the amplifier into a potential oscillator.

**The Solution:** I learned that I could not just force gain through a raw resistor without consequences. I had to decouple the internal gain node from the output port. The solution was designing the active LC L-match network ( $L_2$  and  $C_{out}$ ). This allowed the transistor to see a high impedance (keeping the gain high and stable) while the antenna port still saw a perfectly stable 50  $\Omega$  match. This careful balancing act ultimately pushed  $\mu$  back above 1.00009 across all frequencies.

#### 4.2.2 Challenge 2: The Miller Effect and Reverse Leakage

**The Problem:** During my first attempt at the circuit, I used a standard Common-Source (CS) topology. However, the S-parameter sweeps showed a terrible reverse isolation ( $S_{12}$ ) of only -15 dB. The gate-to-drain parasitic capacitance ( $C_{gd}$ ) was acting as a bridge, leaking the amplified signal backward into the input gate.

**The Solution:** I researched architectural solutions and implemented the Cascode topology. By stacking a Common-Gate transistor ( $M_2$ ) on top of the main amplifying transistor ( $M_1$ ), I effectively pinned the drain voltage of  $M_1$ . This eliminated the voltage swing across  $C_{gd}$ , neutralizing the Miller effect and pushing the reverse isolation down to a highly secure -38.14 dB.

#### 4.2.3 Challenge 3: Parasitic Frequency Shifts

**The Problem:** When calculating the initial value for the gate tuning inductor ( $L_g$ ), I used the standard LC resonance formula ( $f = 1/2\pi\sqrt{LC}$ ) targeting 9.0 GHz. However, upon running the first simulation, the input return loss ( $S_{11}$ ) actually peaked at 8.2 GHz instead of 9.0 GHz. The formula had not accounted for the hidden, non-ideal parasitic capacitances inside the transistor models provided by the 180nm foundry.

**The Solution:** I learned that mathematical formulas only provide a starting point in RF design. To fix the frequency shift, I set up a parametric sweep in Cadence, slowly stepping the value of  $L_g$  down from its calculated ideal value until the resonance shifted

exactly to 9.0 GHz. This taught me the importance of iterative tuning and treating simulator feedback as the ultimate truth.

# Chapter 5

## Conclusion and Future Scope

### 5.1 Conclusion

This summer internship provided an invaluable introduction to the practical realities of Radio Frequency Integrated Circuit (RFIC) design. The primary objective of the project was to design, bias, and optimize a 9.0 GHz Low Noise Amplifier (LNA) using a legacy 180nm technology node. Given the severe high-frequency parasitic limitations of this specific process—where transit frequency ( $f_T$ ) is a significant constraint—achieving high gain while maintaining stability and low noise was a substantial engineering challenge.

Through a structured design approach in Cadence Virtuoso, these challenges were successfully overcome. By implementing an inductively degenerated cascode architecture, the destabilizing Miller effect was neutralized, resulting in excellent reverse isolation. Furthermore, by carefully engineering an active LC L-match network at the output, the circuit was able to maintain a high internal impedance for maximum voltage gain while presenting a perfect  $50\ \Omega$  match to the antenna.

The final simulations achieved a forward gain of 11.06 dB and a Noise Figure of 1.41 dB at 9.0 GHz. Most importantly, Edwards-Sinsky geometric analysis confirmed that the LNA is unconditionally stable across the entire frequency spectrum. This project demonstrates that with precise architectural choices and iterative impedance matching, legacy technology nodes remain highly capable for building sensitive receiver front-ends in radar applications.

### 5.2 Future Scope

While the schematic-level design was successful, transitioning this LNA into a physical, fabricated chip involves several advanced stages:

- **Physical Layout Design:** The next step is drawing the physical layout of the

circuit, ensuring it meets strict Design Rule Checks (DRC) and Layout Versus Schematic (LVS) requirements.

- **Post-Layout Extraction (PEX):** Metal routing introduces unintended parasitic resistances and capacitances. PEX simulation is required to re-verify that the S-parameters and Noise Figure remain within specification after physical wiring is added.
- **System Integration:** The LNA can be integrated with a local oscillator and mixer to create a complete down-converting radar receiver block, suitable for multi-sensor drone detection platforms.

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